Fare Hike, Same Ride

Did Seoul's 2023 and 2025 fare rises push commuters onto 따릉이 public bikes? No, but not for the reason the conventional wisdom gives.

Conventional wisdom: "따릉이 is mainly for leisure and exercise, not a real commuting mode, so subway fare changes will not meaningfully change how it is used." We test the **conclusion** (fares do not matter) and the **premise** (it is a leisure mode) separately.

Design

- **Outcome:** share of each weekday's rides starting in commuting hours (07–09, 18–20). Weather moves the volume of rides, not their timing.
- **Comparison:** same calendar dates one year apart: Oct–Dec 2022 vs 2023 (average fare +18.4%) and Jul–Sep 2024 vs 2025 (+5.2%). One weekday = one observation (59/56 and 62/65 days).
- **Data:** 680,258 station-days of hourly rental records, 2,789 stations, 365 days of 2022 daily totals, KMA daily weather. Zones: 153 subway-exit stations in business districts, 147 park stations.
- **Pre-set rule:** two-sided tests at α = 0.05; a change under 3 percentage points does not count as meaningful.

Findings

1 · Weather drives volume (multiple regression)

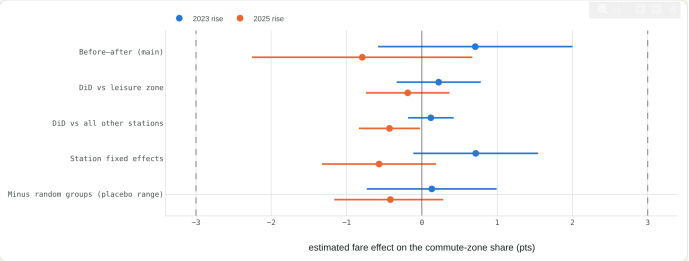
Temperature (cubic), log rain and rest days explain **86%** of daily rentals (held-out 86%), against 46% for temperature alone. Leaving rain out biased the temperature slope down (3,183 → 3,647 rides per °C), because hot days are wetter.

2 · Fares did not move the commuting share

Commute zone, temperature-adjusted: **+0.71 pts** [−0.58, +2.00], p = 0.28 (2023) and **−0.79 pts** [−2.26, +0.67], p = 0.29 (2025). Equivalence tests confirm the change is under 3 points (TOST p = 0.0003, 0.0017). Multiple regression with temperature, rain and weekday gives the same answer (+0.74, −0.81).

3 · The null holds up under causal checks

Difference-in-differences against the leisure zone: +0.22 and −0.19 pts (p = 0.43, 0.51); station fixed effects and a 2,000-draw placebo (p = 0.76, 0.24) agree. Parallel-trend and Chow checks pass.



Every design lands inside ±3 points (dashed). Blue 2023, orange 2025.

4 · The premise fails

Not leisure-led: rest days have **25,726 fewer** rides than workdays at the same weather (F = 13.84, p < 0.001), and park stations are almost as commute-heavy (≈32% vs 33%). But commuting-hour rides are as weather-sensitive as other rides (F = 0.15, p = 0.86): an optional, fair-weather mode.

5 · ML adds a hidden feature

A random forest reaches held-out R^2 0.92 and ranks **hours of rain** second; the fare rise ranks near zero for the share (0.011).

Revised wisdom: 따릉이 is not mainly a leisure toy; a third of weekday rides start in commuting hours. It behaves like a fair-weather option, not a subway substitute, so fare rises leave it alone. **Weather, not fares, moves the bikes.**

Implications

- Do not count on fare rises to shift commuters to public bikes.
- Plan bike supply and rebalancing around weather forecasts (86% of daily demand).
- Measure peak-hour dock availability and Climate Card use, the hidden features most likely to mask a fare effect.

Limitations

- Effects under ~1 point cannot be ruled out; zones are name-based.
- No 2021 placebo year or subway ridership check yet.
- Commuting hours are a proxy; daily rain does not say when it rained.

DATA STORY WEBSITE, CODE REPOSITORY AND RAW DATA: LINKS IN OUR SLACK AND LXP SUBMISSION · AN JAEWON · LEE BOMIN · YUN SANGCHEOL · YANG SHINJE · YOON SOBIN · CHOI BRANDON HYUNGU